

Test yourself

Answers available from twalmsle@usc.edu

Q1: Hoy et al, p56, Q6

Simplify the following expressions

- a) $\frac{\sqrt{a^5}}{a^3}$
- b) $a^2 b^3 / a^2 b$
- c) $b x_1^{b-1} x_2^c / c x_1^b x_2^{c-1}$
- d) $\left(x_1^{1/b} x_2^c\right)^{b/c}$

Q2: Hoy et al, p56, Q1b

Give equations and sketch graphs of the lines passing through (-2,2) and parallel to $y = 2-5x$.

Q3: Hoy et al, p192, Q3

Find the slope of each of the following production functions, $y = f(L)$. Graph the functions and their derivative functions. Give the economics significance of the functions. In each case $L > 0$.

- a) $y = 64 \cdot L^{1/4}$
- b) $y = 10L + 2L^{1/2}$
- c) $y = 5L^3$

Q4: Hoy et al, p56, Q2

In a class of 120 students, everyone would take two hamburgers if the price were zero and no one would buy hamburgers if the price were \$4 or more. Assume that the class demand curve is linear and give the equation. Explain what this implies about the demand for hamburgers where the price is \$3.99?

Q5: Hoy et al, p56, Q4

Total revenue is price x quantity sold. Show that the total revenue curve corresponding to the demand curve found in the previous question is a quadratic. Is it convex or concave? At what value does its maximum or minimum occur?

Q6: Hoy et al, p230, Q5

A profit maximizing firm has a total cost function

$$C = 0.5x^3 - 3x^2 + 6x + 50$$

And sells into a competitive market on which the price is \$1. What output should it produce? What would your answer be if the price were \$2?

Q7: Hoy et al, p464, Q4

Show that the production function $y = x_1^{1/2} \cdot x_2^{2/3}$ is homogenous to degree 7/6.

Q8: Hoy et al, p464, Q5

For the Cobb Douglas production function $y = A \cdot x_1^a \cdot x_2^b$, $A > 0$ and $a, b > 0$ defined on $\mathbb{R}_{++}^2$, show that the Euler theorem applies so that:

$$f_1 x_1 + f_2 x_2 = k \cdot f(x_1, x_2)$$

Where k is the degree of homogeneity.

Q9: Hoy et al, p515, Q1

Solve the following constrained optimization and minimization problems, illustrating your answers with diagrams in each case. Confirm that each is an optimum:

- a) Max $y = 2x_1 + 3x_2$ subject to $2x_1^2 + 5x_2^2 = 10$
- b) Max $y = x_1^{0.25} \cdot x_2^{0.75}$ subject to $2x_1^2 + 5x_2^2 = 10$
- c) Min $y = 2x_1 + 4x_2$ subject to $x_1^{0.25} \cdot x_2^{0.75} = 10$

Q10

Show that the price elasticity of demand is equal to 1 for a Cobb Douglas demand function.